OData: overview for implementers

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Title

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From Version

Up to Version

Addressees

Context

Data stored in CCH® Tagetik can be consumed with CCH® Tagetik standard reporting or be analyzed using a third-party BI tool. When CCH® Tagetik is deployed on cloud it is necessary to expose the data by means of RESTful API and OData service to allow third-party BI tool to read it and display it.

Description

Glossary

RESTful API: application program interface (API) that uses HTTP requests to access and use data.
OData: Open Data Protocol used for building and consuming RESTful APIs.
OASIS standard: It is a Protocol developed by the non-profit Organization for the Advancement of Structured Information Standards.
JSON file: A JSON file is a file that stores simple data structures and objects in JavaScript Object Notation (JSON) format, which is a standard data interchange format. It is primarily used for transmitting data between a web application and a server.
SAML: Security Assertion Markup Language (SAML) is an open standard that allows identity providers (IdP) to pass authorization credentials to service providers (SP).
SCIM: System for Cross-domain Identity Management (SCIM) is a standard for automating the exchange of user identity information between identity domains, or IT systems.
Open API: is an Interface Description Language for describing RESTful APIs expressed using JSON.

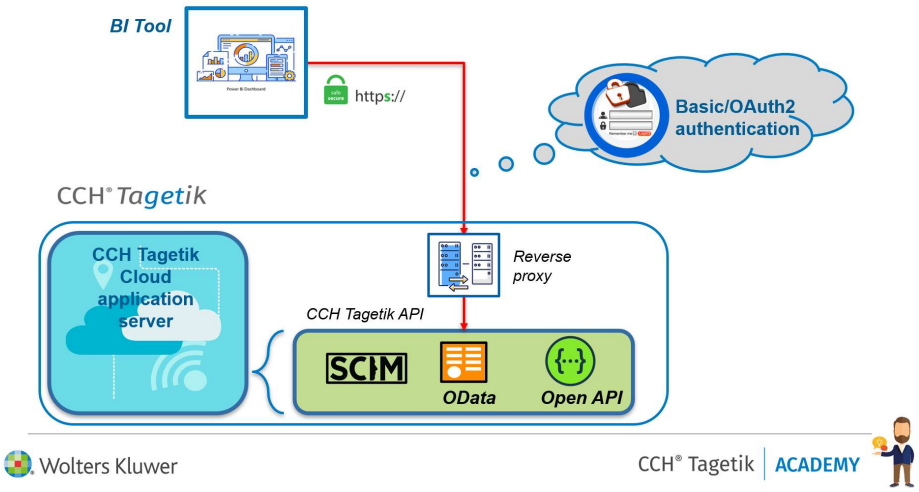
RESTful API and OData service

A RESTful API is an architectural style for an application program interface (API) that uses HTTP requests to access and use data. OData (Open Data Protocol) is an ISO/IEC approved, OASIS standard that defines a set of best practices for building and consuming RESTful APIs. CCH® Tagetik exposes its data by RESTful APIs using in OData protocol. At present the OData service used is version 4. It means that the CCH® Tagetik APIs generate a JSON file that the browser translates into an XML.

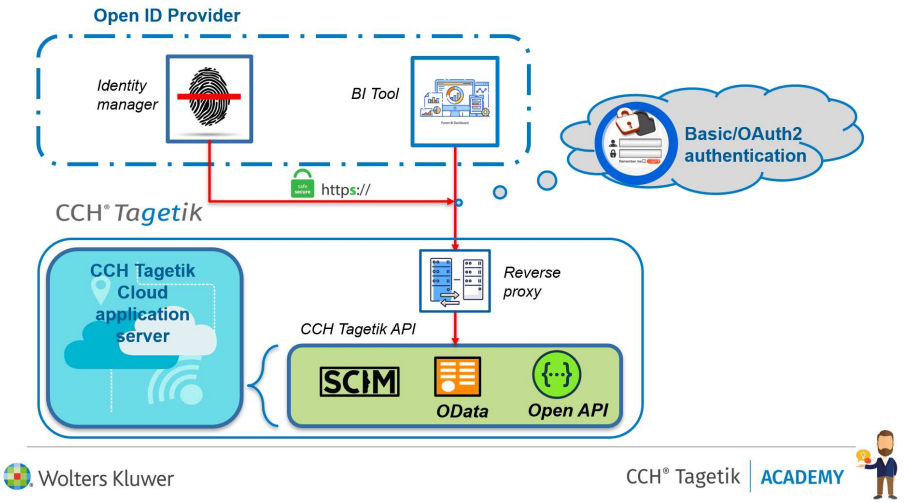
Why is OData the solution?

CCH® Tagetik allows to expose its data by RESTful APIs using the OData protocol. OData eases the access of data from external connectors without a direct connection to the database. Consumers can access CCH® Tagetik data directly via HTTPs requests.

Typical frameworks



The image above represents the most typical connection's framework. The BI tool is standalone, and it is not installed in an External ID provider. CCH® Tagetik data on cloud is exposed by the different RESTful API. SCIM represents information for user provisioning, Open API is a standard for metadata whereas the OData standard provides the real Data of CCH® Tagetik. The BI tool pulls its requests to Tagetik, and the reverse proxy allows the whitelisted IP to consume the data exposed by API. The user will be authenticated by a Basic authentication or by an OAuth2 authentication.



In this second framework the BI tool is installed on an Open ID provider (could be for example Azure). Using this solution, the preferred user authentication is OAuth2 because it provides a stronger security for the data. However, the use of basic authentication is allowed too. The Identity manager of the open ID provider could also consume SCIM for user provisioning. In both solutions all data flows are encrypted using HTTPS protocol.

Implementer skills required

- CCH® Tagetik Repository: basic knowledge
- Analytic Information Hub: advanced knowledge
- Network Administration: basic knowledge

Implementation of an OData connection

- OData: General settings and Users setup (/CustomerPortal/s/article/OData-General-settings-and-Users-setup)
- How to setup endpoint authentication and network security in OData service framework (/CustomerPortal/s/article/How-to-setup-endpoint-authentication-and-network-security-in-OData-service-framework)

Responsibilities of OData connection implementation

Implementing OData connection is a joint effort of different actors involved:

- Implementer
  - Supervise the project and assign the different responsibilities. In some case the implementer has the duties of System administrator.
- System administrator:
  - REST APIs activation.
  - Setup users and roles.
- Cloud team:
  - Setup of proxy's whitelist.
  - Setup of file standalone.xml.
  - In case of OAuth2 authentication: Connection/Authentication with Open IP provider.
- Customer IT:
  - In case of OAuth2 authentication configuration of Open IP provider.
  - Provide the list of the IPs to be whitelisted.

Loading OData to CCH® Tagetik application

There are two ways to load OData to CCH® Tagetik: by ETL or by Data source mapping. The following articles explain how to implement the two components in order to extract OData from an external source and load them to CCH® Tagetik.

- How to implement and run an ETL routine to extract OData service (/CustomerPortal/s/article/How-to-implement-and-run-an-ETL-routine-to-extract-OData-service)
- How to implement a Datasource mapping to load OData to AW (/CustomerPortal/s/article/How-to-implement-a-Datasource-mapping-to-load-OData-to-AW)

Example

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